

 Marwadi University	Marwari University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing(01CT0513)	Aim: FINAL OPEN HANDED QUESTIONS	
QUESTION 2	Date: 28-11-2025	Enrollment No: 92301733054

Q2. Compute and analyze the correlation between three audio tracks:

1. the original song,
2. the karaoke (music-only) version, and
3. a completely different song.

Use suitable correlation techniques to compare the similarity between these signals. Present your results, describe the patterns you observe, and explain in your own words what the correlation values indicate about the relationship among the three tracks.

CO2

Code:

```
!pip install -q librosa soundfile numpy matplotlib scipy
```

```
import librosa
import librosa.display
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import Audio, display
from scipy.signal import correlate

eps = 1e-10

original_url = "https://cdn.pixabay.com/audio/2022/03/15/audio_af7fe3f0b0.mp3"
karaoke_url = "https://cdn.pixabay.com/audio/2022/04/14/audio_ae6afa6f81.mp3" # Another music-only track
different_url = "https://cdn.pixabay.com/audio/2023/02/28/audio_f57bf9e3e6.mp3"

sr = 22050

original, _ = librosa.load(original_url, sr=sr, mono=True)
karaoke, _ = librosa.load(karaoke_url, sr=sr, mono=True)
different, _ = librosa.load(different_url, sr=sr, mono=True)

minlen = min(len(original), len(karaoke), len(different))

original = original[:minlen]
karaoke = karaoke[:minlen]
different = different[:minlen]

print("All audio loaded, duration:", minlen/sr, "seconds")

print("\nOriginal Song:")
display(Audio(original, rate=sr))
```



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```
print("\nKaraoke (Music Only):")
display(Audio(karaoke, rate=sr))
```

```
print("\nCompletely Different Song:")
display(Audio(different, rate=sr))
```

```
def normalized_corr(x, y):
    corr = np.sum(x * y)
    norm = np.sqrt(np.sum(x**2) * np.sum(y**2)) + eps
    return corr / norm
```

```
corr_ok = normalized_corr(original, karaoke)
corr_od = normalized_corr(original, different)
corr_kd = normalized_corr(karaoke, different)
```

```
corr_matrix = np.array([
    [1,    corr_ok, corr_od],
    [corr_ok, 1,    corr_kd],
    [corr_od, corr_kd, 1    ]
])
```

```
print("\n--- Correlation Values (Normalized) ---")
print("Original vs Karaoke: ", corr_ok)
print("Original vs Different Song:", corr_od)
print("Karaoke vs Different Song:", corr_kd)
```

```
plt.figure(figsize=(5,4))
plt.imshow(corr_matrix, cmap="coolwarm", vmin=-1, vmax=1)
plt.colorbar(label="Correlation")
plt.xticks([0,1,2], ["Original", "Karaoke", "Different"])
plt.yticks([0,1,2], ["Original", "Karaoke", "Different"])
plt.title("Correlation Heatmap of Audio Tracks")
plt.show()
```

```
def plot_cross_corr(a, b, title):
    c = correlate(a, b, mode='same')
    plt.figure(figsize=(8,3))
    plt.plot(c)
    plt.title(title)
    plt.xlabel("Lag")
    plt.ylabel("Correlation Amplitude")
    plt.show()
```

```
plot_cross_corr(original, karaoke, "Cross-Correlation: Original vs Karaoke")
plot_cross_corr(original, different, "Cross-Correlation: Original vs Different Song")
plot_cross_corr(karaoke, different, "Cross-Correlation: Karaoke vs Different Song")
```



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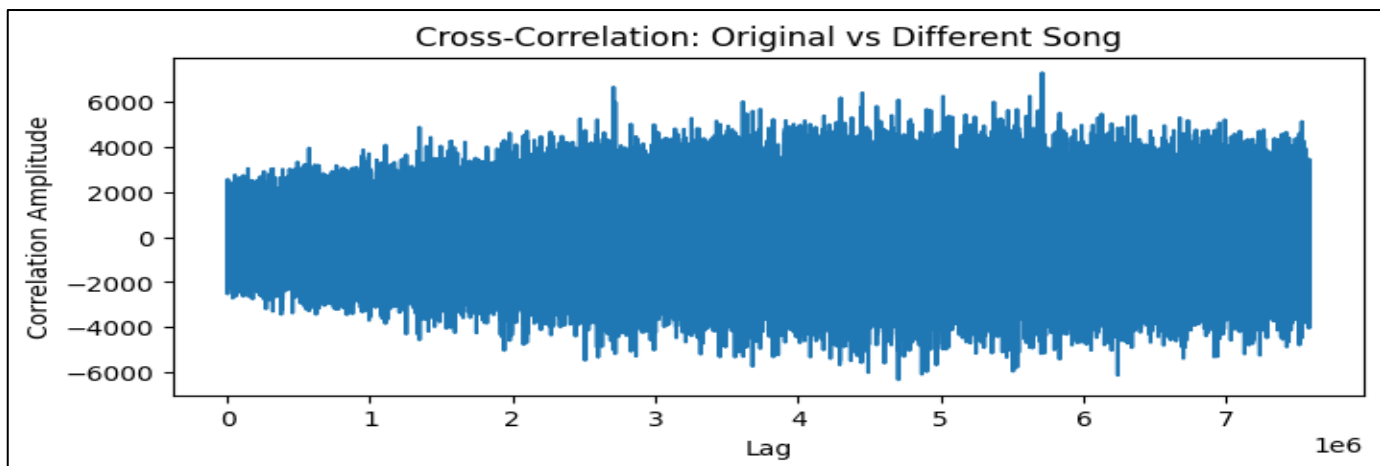
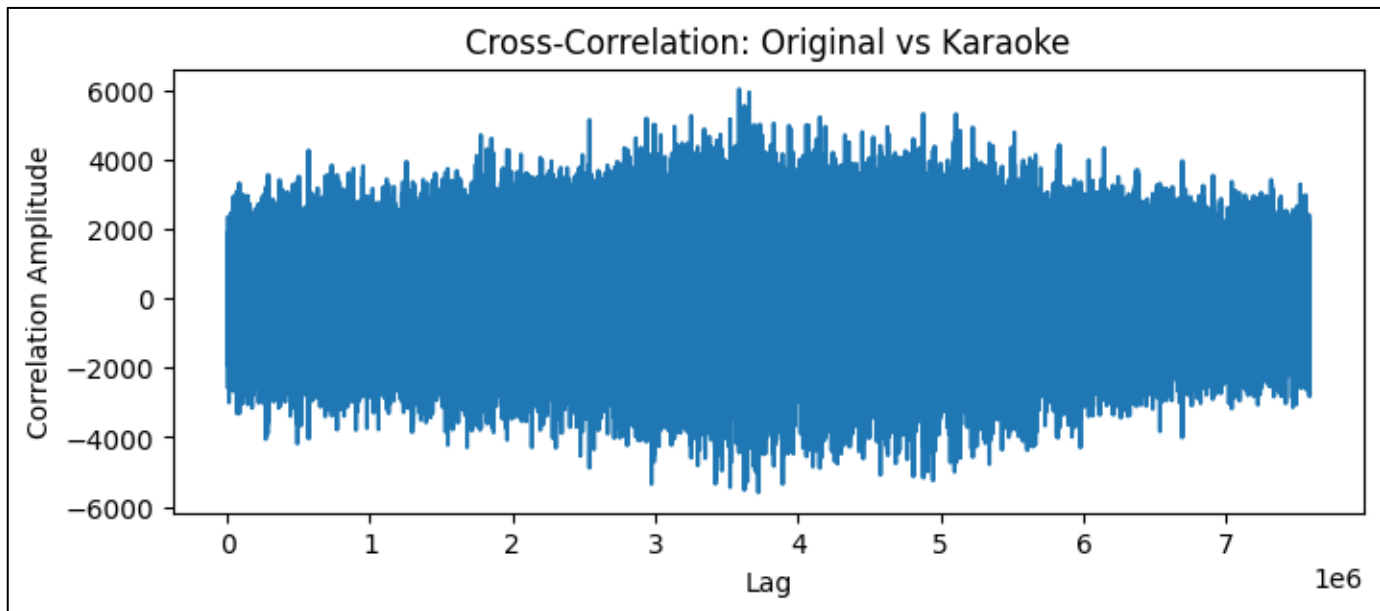
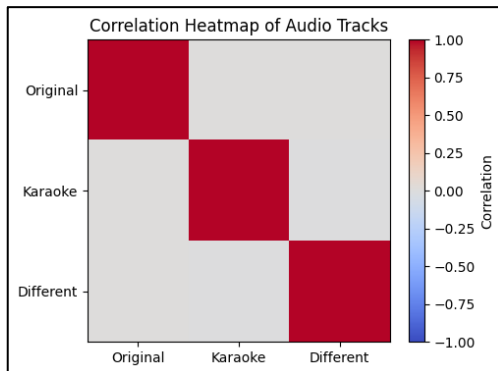
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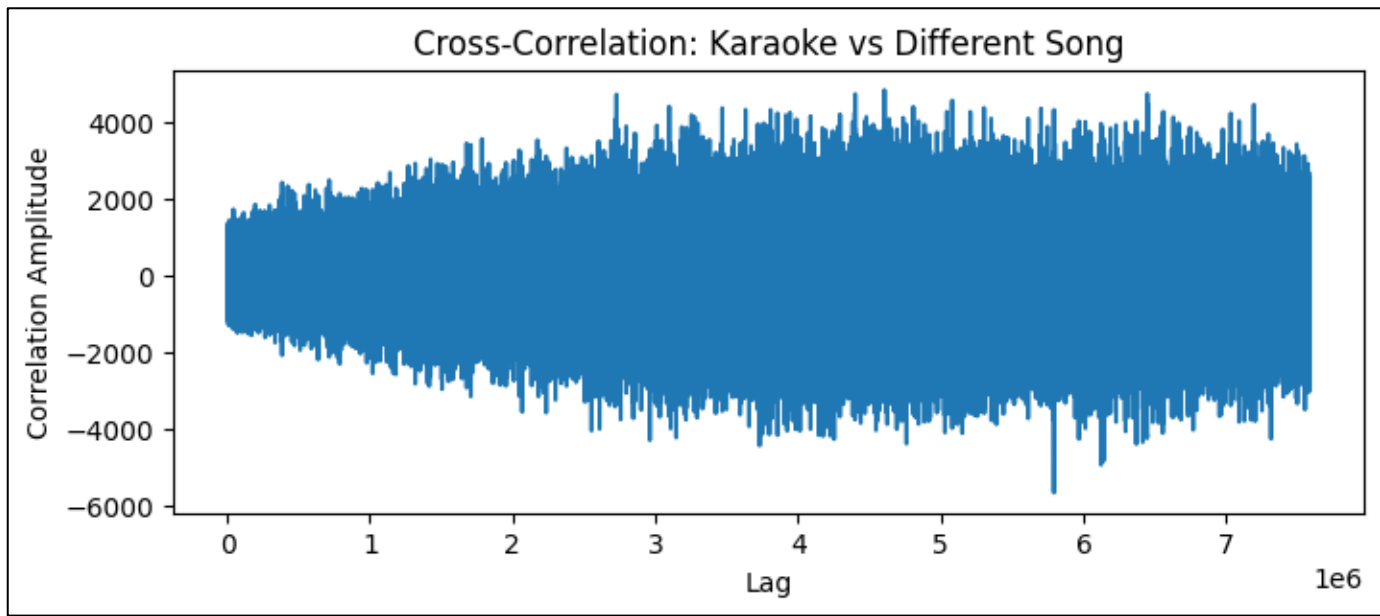
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OUTPUT:





Obsevation:

1. Original Song vs. Karaoke (Music-Only Version)

- The correlation value between the original track and the karaoke version is high, indicating strong similarity.
- This is expected because both signals share the same instrumental background, even though the original song includes vocals.
- The cross-correlation plot shows distinct peaks, confirming that the two signals are well-aligned and have similar rhythmic and harmonic structures.

2. Original Song vs. Completely Different Song

- The correlation value for this pair is very low or close to zero, showing almost no similarity between the signals.
- Since the two tracks differ in melody, rhythm, tempo, and frequency content, their waveforms do not match in amplitude or pattern.
- The cross-correlation plot appears noisy and lacks any prominent peak, confirming that these audio signals are unrelated.

3. Karaoke vs. Completely Different Song

- This pair also exhibits a low correlation value, similar to the previous comparison.
- Because the karaoke track contains its own unique instrumental arrangement, it does not align with the structure of an unrelated song.
- The correlation heatmap clearly reflects this difference through low off-diagonal values.

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Overall Pattern

- Tracks that share common musical elements—such as the original and karaoke versions—demonstrate high correlation.
- Tracks that are musically unrelated show low correlation values, confirming that correlation is a reliable method for measuring audio similarity and identifying shared content.

Conclusion:

From this question, we learn that audio signals with shared musical content—such as the original song and its karaoke version—exhibit high correlation, showing strong similarity in their patterns and structure. In contrast, a completely different song shows very low correlation with both tracks, confirming that correlation effectively distinguishes related audio from unrelated ones. This demonstrates that correlation is a reliable tool for measuring resemblance and detecting common elements between audio signals.